

Maritime Oil-Spill Attribution Intelligence

Scientific Architecture, Hudrodynamic Modeling & Forensic Methodologies

Classification: Technical Dossier / Decision Support	Version: 1.0 (Production Release)
Problem Statement: SIH26143 - Spatiotemporal Attribution	Date: 2026-09-05 17:18 UTC
Core Paradigm: Lagrangian Inverse Backtracking	Application: Decision Support & Maritime Forensics

1. Executive Overview & Problem Formulation

Satellite surveillance detects oil slicks only after they have surfaced and spread across the ocean surface. By the time a satellite observes the slick, ocean currents and winds have transported, deformed, and dispersed the oil miles away from its release location, while the responsible vessel has steamed outside the satellite swath. This platform resolves the **Spatiotemporal Inverse Attribution Problem**: backtracking an observed synthetic aperture radar (SAR) slick through historical space and time, deriving the origin envelope and release window, and correlating candidate vessel AIS transponder tracks.

2. End-to-End Forensic Pipeline Architecture

Stage	Operational Module	Inputs & Telemetry	Analytical Output
1	SAR Detection	Sentinel-1 C-band SAR (OData / GeoTIFF)	Adaptive CFAR segmentation, slick polygon, centroid
2	Lookalike Screening	Morphology + Open-Meteo wind speed	Low-wind calms / biogenic slick rejection audit
3	Lagrangian Drift	Copernicus Marine CMEMS + 10m Wind	Backward Monte Carlo particle swarm advection
4	Origin Estimation	Particle ensemble spatial density	Origin centroid, dispersion cloud covariance
5	Release Windowing	Fay weathering spreading kinetics	Spill release window [T_min, T_max]
6	AIS Interception	Historical/Live AIS GPS streams	Waypoint interpolation, AIS blackout gap detection
7	Attribution Scoring	Spatiotemporal distance matrices	Ranked candidate leaderboard & forensic dossier

3. SAR Slick Detection & Lookalike Discrimination

Physical Principle: Hydrocarbon films dampen capillary and short gravity waves on the sea surface, dramatically suppressing radar backscatter. Consequently, in Sentinel-1 Synthetic Aperture Radar (SAR) backscatter imagery, oil slicks appear as dark anomalies contrasted against the background ocean.

Constant False Alarm Rate (CFAR) Adaptive Thresholding:

To handle varying sea states across wide satellite swaths, CFAR dynamically calculates local thresholds:

$$T(x, y) = \mu_b(x, y) - k \cdot \sigma_b(x, y)$$

where μ_b is the local background mean backscatter, σ_b is the background standard deviation, and k is a sensitivity multiplier (typically 2.0 to 3.0).

Feature Extraction & Lookalike Rejection:

- Elongation Ratio:** $E = L_{major} / L_{minor}$. Slicks resulting from moving vessels exhibit high elongation ($E \geq 3.0$).
- Isoperimetric Quotient (Circularity):** $Q = 4\pi A / P^2$. Rejects irregular natural calm areas and broad biogenic slicks.
- Wind-Speed Thresholding:** Slicks observed in wind regimes below 2.0 m/s (natural calm lookalikes) or above 12.0 m/s (high dispersion) are rejected.

4. Backward Lagrangian Hydrodynamic Drift Engine

Following slick detection at satellite acquisition time T_{SAR} , the engine initializes a Monte Carlo particle swarm ($N = 1,000$ to $5,000$ virtual particles) across the segmented polygon. The ensemble is integrated backwards in time with negative time increments ($\Delta t = -30$ min):

Governing Advection-Diffusion Equation:

$$\mathbf{x}_i(t - \Delta t) = \mathbf{x}_i(t) - [\alpha \cdot \mathbf{u}_{current}(\mathbf{x}, t) + \gamma \cdot \mathbf{R}(\theta) \cdot \mathbf{w}_{10}(\mathbf{x}, t)] \Delta t + \eta_i(t)$$

- **Ocean Current Advection ($\mathbf{u}_{current}$):** Copernicus Marine (CMEMS / HYCOM) surface velocity vectors. Scaling coefficient $\alpha = 1.00$.
- **Atmospheric Wind Leeway ($\mathbf{w}_{10}$):** Open-Meteo Marine Global NWP 10-meter wind vectors. Empirical leeway factor $\gamma = 0.03$ (3.0%).
- **Coriolis Leeway Deflection ($\mathbf{R}(\theta)$):** Rotational deflection angle θ (Ekman spiral) accounted for according to hemisphere.
- **Turbulent Eddy Diffusion (η_i):** Stochastic Brownian perturbation modeling ocean sub-grid turbulence:

$$\Delta \mathbf{x}_{turbulent} = \mathbf{R}_{norm} \cdot \text{sqrt}(2 \cdot \mathbf{K}_h \cdot \Delta t)$$

where $\mathbf{K}_h$ is the horizontal eddy diffusivity (2.5 to 10.0 m^2/s) and $\mathbf{R}_{norm}$ is a standard Gaussian random vector $N(0, 1)$ seeded deterministically.

5. Spatiotemporal Uncertainty Envelope & Release Window

The platform avoids static coordinates, expressing the origin as a **Probability Dispersion Cloud** at each historical timestep;

$$\sigma = \text{sqrt}((1 / N) \sum || \mathbf{x}_i - \mu ||^2)$$

Release Window Estimation: Based on Fay's empirical three-phase spreading kinetics and weathering models (evaporation and emulsification), the physical age of the slick is constrained to an interval $[T_{spill,min}, T_{spill,max}]$, establishing the time window for vessel intersection.

6. AIS Vessel Track Interception & Gap Analysis

AIS transponder telemetry within the spatial-temporal envelope is ingested and interpolated along spherical geodesic tracks using the Great-Circle Haversine formula:

$$d = 2R \cdot \arcsin(\text{sqrt}(\sin^2(\Delta\phi/2) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2(\Delta\lambda/2)))$$

Dark Vessel & AIS Gap Detection: Transponder blackouts exceeding 45 minutes across the spill origin corridor are flagged as anomaly gaps (Δt_{gap}), penalizing the vessel's continuity score.

7. Multi-Factor Attribution Scoring Matrix

Attribution is synthesized into a calibrated composite score (0 to 100 points) evaluating four independent forensic pillars:

$$\text{Attribution Score} = w_1 \cdot S_{\text{proximity}} + w_2 \cdot S_{\text{temporal}} + w_3 \cdot S_{\text{relevance}} + w_4 \cdot S_{\text{continuity}}$$

Factor	Weight	Formula / Definition	Forensic Rationale
Spatial Proximity (S_prox)	35%	$S_{\text{prox}} = \max(0, 100 \cdot (1 - d_{\text{min}} / R_{\text{threshold}}))$ where $R_{\text{threshold}} = 25 \text{ km}$	Closest point of approach (CPA) between vessel trajectory and backward drift probability centroid.
Temporal Coincidence (S_time)	30%	$S_{\text{time}} = 100 \cdot \exp(-(\Delta t_{\text{passage}} - t_{\text{release}})^2 / (2 \cdot \sigma_t^2))$	Evaluates overlap between vessel transit time and the scientifically derived release window.
Vessel Relevance (S_vessel)	20%	Crude/Product Tanker: 95-100 Chemical/Bulk Carrier: 70-85 Cargo/Container: 50-65 Fishing/Pleasure: 10-25	Risk classification based on cargo carrying capacity, bunker fuel type, and historical spill potential.
AIS Continuity (S_cont)	15%	Continuous transmission: 100 Minor telemetry jitter (<45m): 85 Unexplained gap across spill origin: 20	Penalizes deliberate transponder shutdowns, track spoofing, or irregular maneuvers near origin.

8. Forensic Integrity & Legal Safeguards

To guarantee admissibility and evidentiary defensibility before maritime tribunals and international bodies:

- **Non-Accusatory Wording:** The platform never issues automated verdicts of legal guilt. Reports state: *"Candidate vessel [Name] (MMSI: [X]) is the highest-ranked candidate based on available spatiotemporal, hydrodynamic, and AIS evidence."*
- **Strict Anti-Fabrication Rule:** When external sensor providers are unconfigured, the system reports UNCONFIGURED honestly. Synthetic fallbacks in production are strictly forbidden.
- **Immutable Provenance Audit Trail:** Every analysis step records sensor platform IDs, acquisition timestamps, grid parameters, and SHA-256 hashes of input rasters in an immutable audit chain.
- **Decision-Support Dossier:** Results compile directly into exportable technical dossiers embedding cartographic overlays, score breakdowns, lookalike rejection logs, and sensor limitation disclosures.